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Tether traction systems and methods of use thereof

Abstract

The present disclosure relates generally to the field of medical devices. In particular, the present disclosure relates to traction systems, and methods of use thereof, for endoscopic procedures such as tissue dissection. For example, a traction system may include a filament extendable along an outer surface of an endoscope with a distal end of the filament attachable to a medical device engaged with a target tissue of a body lumen.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of and claims the benefit of the earlier filing date of U.S. patent application Ser. No. 17/940,676, filed Sep. 8, 2022, which claims the benefit of U.S. patent application Ser. No. 15/930,607, filed on May 13, 2020, which claims the benefit of priority under 35 U.S.C. § 119 to U.S. Provisional Patent Application No. 62/861,555, filed on Jun. 14, 2019, all of which applications are incorporated herein by reference in their entireties for all purposes. Any and all priority claims identified in the Application Data Sheet, or any correction thereto, are hereby incorporated by reference under 37 C.F.R. § 1.57.

FIELD

(1) The present disclosure relates generally to the field of medical devices. In particular, the present disclosure relates to traction systems, and methods of use thereof, for endoscopic procedures such as tissue dissection.

BACKGROUND

(2) Performing an endoscopic tissue resection/dissection procedure may include maintaining traction as the boundaries of the target tissue are dissected for accuracy and efficiency of the procedure. Physicians may use a tethered system in which an endoscopic clip is attached to a length of filament extending external to the patient to retract/immobilize a target tissue for dissection along the tissue margins and/or to retrieve the dissected target tissue for biopsy.

(3) It is with the above considerations in mind that the improvements of the present disclosure may be useful.

SUMMARY

(4) In one aspect, the present disclosure relates to a system for use with an endoscope comprising a

cap attachable to (or a portion of) a distal end of an endoscope having a working channel extending therethrough. The cap may include a lumen coextensive with the working channel. A first guide may be attached to/mounted on an outer surface of the cap. A first filament may be extendable through the first guide and along an outer surface of the endoscope and the cap. A first endoscopic instrument may be disposable within the working channel. A distal end of the first filament may be attached to a distal end of the first endoscopic instrument.

(5) In the described and other embodiments, the first endoscopic instrument may include a sheath and a first medical device may be releasably attached to a distal end of the sheath. The first medical device may include a tissue clamp. The distal end of the first filament may be attached to an arm of the tissue clamp. A handle may be operatively attached to a proximal end of the first endoscopic instrument. The handle may be configured to move the first medical device between a first and second position. The handle may be configured to release the first medical device from the distal end of the sheath. A flexible elongate shaft may be extendable along the outer surface of the endoscope and the cap and through the first guide. The first filament may extend through the flexible elongate shaft. A handle assembly may be attachable to a proximal end of the flexible elongate shaft. The handle assembly may include first, second, and third gripping elements. The first gripping element may be configured to advance and retract the flexible elongate shaft through the first guide. The second and third gripping elements may be attached to each other by a connector. A proximal end of the first filament may be attached to the connector.

(6) In another aspect, the present disclosure relates to a system for use with an endoscope comprising a cap attachable to (or a portion of) a distal end of an endoscope having a working channel extending therethrough. The cap may include a lumen coextensive with the working channel. A first guide and a second guide may be attached to/mounted on an outer surface of the cap. A first filament may be extendable through the first guide and along an outer surface of the endoscope and the cap. A first endoscopic instrument may be disposable within the working channel. A distal end of the first filament may be attached to a distal end of the first endoscopic instrument.

(7) In the described and other embodiments, the first endoscopic instrument may include a sheath and a first medical device may be releasably attached to a distal end of the sheath. The first medical device may include a tissue clamp. The distal end of the first filament may be attached to an arm of the tissue clamp. A second filament may extend through the second guide and along an outer surface of the endoscope and the cap. A distal end of the second filament may be attached to an arm of a second medical device. A flexible elongate shaft may extend along the outer surface of the endoscope and the cap and through the first guide. The first filament may extend through the flexible elongate shaft. A flexible elongate shaft may be extendable along the outer surface of the endoscope and the cap and through the second guide. The second filament may extend through the flexible elongate shaft.

(8) In another aspect, the present disclosure relates to a system for performing an endoscopic procedure. The system may comprise a first guide extendable along an outer surface of the endoscope. The guide may be configured for mounting on an endoscope or a portion of an endoscope (part of the endoscope or separately formed and attached to/mounted on the endoscope). A first filament may be extendable through the first guide and along an outer surface of the endoscope; and a first endoscopic instrument may be disposable within the working channel, wherein a distal end of the first filament may be attached to a distal end of the first endoscopic instrument.

(9) In yet another aspect, the present disclosure relates to a method comprising advancing a first instrument through a working channel of an elongate tubular member and through a lumen of a cap attached to/mounted on a distal end of the elongate tubular member such that a distal end of the first instrument may extend distally beyond a distal end of the cap.

(10) In the described and other embodiments, a medical device attached to the distal end of the first

instrument may move from a closed position to an open position. A distal end of a filament extending along an outer surface of the elongate tubular member and, optionally, cap may be attached to an arm of the medical device. The filament may extend through a guide attached to/mounted on an outer surface of the elongate tubular member or cap. The medical device may be moved from the open position to the closed position. The first instrument may be retracted such that the medical device may be disposed within the working channel of the elongate tubular member. The elongate tubular member may be advanced through a body lumen to position the distal end of the elongate tubular member or cap adjacent to a target tissue. The medical device may be advanced distally beyond a distal end of the elongate tubular member or cap. The medical device may be moved from the closed position to the open position. The medical device may be moved from the open position to the closed position to engage the target tissue. The medical device may be disengaged from the distal end of the first instrument. A proximal end of the filament may be proximally retracted to apply tension to the target tissue. The first instrument may be replaced with a second instrument within the working channel of the elongate tubular member. The target tissue may be manipulated with a medical device attached to a distal end of the second instrument.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Non-limiting embodiments of the present disclosure are described by way of example with reference to the accompanying figures, which are schematic and not intended to be drawn to scale. In the figures, each identical or nearly identical component illustrated is typically represented by a single numeral. For purposes of clarity, not every component is labeled in every figure, nor is every component of each embodiment shown where illustration is not necessary to allow those of ordinary skill in the art to understand the disclosure. In the figures:

- (2) FIGS. 1A-1C provide perspective views of an instrument, according to one embodiment of the present disclosure.
- (3) FIG. 2 provides a perspective view of a conventional tether traction system.
- (4) FIG. 3 provides a perspective view of a tether traction system, according to one embodiment of the present disclosure.
- (5) FIG. 4 provides a perspective view of a filament loop disposed around an arm of a medical device, according to one embodiment of the present disclosure.
- (6) FIG. 5 provides a perspective view of a tether traction system, according to one embodiment of the present disclosure.
- (7) FIG. 6 provides a perspective view of a tether traction system, according to one embodiment of the present disclosure.
- (8) FIG. 7 provides a perspective view of a handle assembly of a flexible elongate shaft, according to one embodiment of the present disclosure.
- (9) FIG. 8 provides a perspective view of a handle assembly of a flexible elongate shaft, according to one embodiment of the present disclosure.

DETAILED DESCRIPTION

(10) The present disclosure is not limited to the particular embodiments described herein. The terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting beyond the scope of the appended claims. Unless otherwise defined, all technical terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the disclosure belongs.

(11) Although embodiments of the present disclosure are described with specific reference to endoscopic systems and methods designed to provide traction within the gastrointestinal tract during a tissue dissection/resection procedure, it should be appreciated that such systems and

methods may be used to manipulate a variety of tissues within a variety of different body lumens and/or body passages in conjunction with or independent of an endoscope.

(12) As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” or “includes” and/or “including” when used herein, specify the presence of stated features, regions, steps elements and/or components, but do not preclude the presence or addition of one or more other features, regions, integers, steps, operations, elements, components and/or groups thereof.

(13) As used herein, the term “distal” refers to the end farthest away from the medical professional when introducing a device into a patient, while the term “proximal” refers to the end closest to the medical professional when introducing a device into a patient.

(14) Referring to FIGS. 1A-1C, in one embodiment, a first endoscopic instrument **140** for use with an endoscopic system **100**, **200**, **300**, **400** of the present disclosure may include a handle **144** operatively attached to a proximal end of a flexible elongate sheath **142** (e.g., catheter, etc.) and a first medical device **146** releasably attached to a distal end of the flexible elongate sheath **142**. In one embodiment, the first medical device **146** may include a tissue clamp or clip comprising arms or jaws **146a**, **146b** configured to move upon actuation of the handle **144** between a first (e.g., open) position (FIG. 1A) and a second (e.g., closed) position (FIG. 1B). In various embodiments, in the second position the arms or jaws **146a**, **146b** of the first medical device **146** may engage a target tissue **148** within a body lumen, e.g., a target tissue of the gastrointestinal (GI) tract. In various additional embodiments, the handle **144** may be further actuated to disengage (e.g., release) the first medical device **146** from the distal end of the sheath **142** (FIG. 1C) such that the first medical device **146** may remain engaged with the target tissue **148** after the first endoscopic instrument **140** has been removed from within the patient.

(15) Referring to FIG. 2, in one embodiment, an endoscopic system **100** may include a cap **120** (e.g., endoscopic cap, etc.) attachable (e.g., by a friction or interference fit, a threaded fit, a snap-lock fit, etc.) to the distal end **112** of a flexible elongate tubular member **110** (e.g., endoscope, etc.) having a proximal end (not shown), a distal end **112**, and a lumen or working channel **114** extending therebetween. A lumen **124** defined by the cap **120** may be coextensive with (e.g., aligned with) the working channel **114**. A first endoscopic instrument **140** may be movably/slidably disposable within the working channel **114** and lumen **124** such that a first medical device **146** attached to a distal end of a sheath **142** of the first endoscopic instrument **140** may be extendable distally beyond the distal end of the cap **120**, e.g., to engage a target tissue within a body lumen. A first filament **130** (e.g., suture, wire, string, etc.) may be extendable along an outer surface of the elongate tubular member **110** and an outer surface of the cap **120** and a distal end **132** of the first filament **130** may be attached to (e.g., tied, etc.) a portion of the first medical device **146**, including, for example an arm **146a** of the first medical device **146**. In various embodiments, as discussed above, the first medical device **146** may be movable from the first to second positions to engage a target tissue with the first and second arms **146a**, **146b** and the first medical device **146** to be disengaged/released from the distal end of the sheath **142**. A proximal end **136** of the first filament **130** may be extendable beyond the proximal end of the elongate tubular member **110** and external to the patient such that a medical professional may proximally retract the first filament **130** to apply tension/traction to the target tissue engaged with the first medical device **146**.

(16) Referring to FIG. 3, in one embodiment, an endoscopic system **200** of the present disclosure may include a cap **220** (e.g., endoscopic cap, etc.) attachable to (e.g., by a friction or interference fit, a threaded fit, a snap-lock fit, etc.) the distal end **112** of a flexible elongate tubular member **210** (e.g., endoscope, etc.) having a proximal end (not shown), a distal end **212** and a working channel **214** extending therebetween. A lumen **224** defined by the cap **220** may be coextensive with (e.g., aligned with) the working channel **214**. A first guide **226** (e.g., filament guide, suture guide, tubular support, ring, etc.) defining an open channel therethrough may be attached to/mounted on an outer

surface of the cap **220**. A first filament **230** (e.g., suture, wire, string, or other flexible material, etc.) may be extendable along an outer surface of the elongate tubular member **210** and an outer surface of the cap **220** and through the channel of the first guide **226**.

(17) As discussed above, a first endoscopic instrument **140** may be movably/slidably disposable within the working channel **214** and lumen **224** such that a first medical device **146** attached to a distal end of the first endoscopic instrument **140** may be extendable distally beyond the distal end of the cap **220**, e.g., to engage a target tissue. A distal end **232** of the first filament **230** may be attached (e.g., tied, etc.) to an arm **146a** of the first medical device **146**. The first medical device **146** may then be moved from the first to second positions to engage a target tissue **148** and the first medical device **146** disengaged from the distal end of the sheath **142**.

(18) A proximal end **236** of the first filament **230** may extend beyond the proximal end of the elongate tubular member **210** and external to the patient such that a medical professional may proximally retract the first filament **230** to apply tension/traction to the target tissue **148** engaged with the first medical device **146** along a longitudinal axis of the cap **220** through the first guide **226**.

(19) In use, and by way of example, an endoscopic system **200** of the present disclosure may be assembled (e.g., prior to insertion into the patient) by attaching the cap **220** to the distal end **212** of the elongate tubular member **210**. The first endoscopic instrument **140** may then be advanced through the working channel **214** of the elongate tubular member **210** and through the lumen **224** of the cap **220** such that the first medical device **146** attached to the distal end of the sheath **142** of the first endoscopic instrument **142** may extend distally beyond a distal end of the cap **220**. The handle **144** of the first endoscopic instrument may then be actuated to move the first medical device **146** from the second position (FIG. 1B) to the first position (FIG. 1A), e.g., such that the arms **146a**, **146b** are open/separated in a Y-shape. The first filament **230** may then be advanced through the open channel of the first guide **226** of the cap **220** and a loop **232** formed from a distal portion of the first filament **230** may be disposed around one of the arms **146a**, **146b** of the first medical device **146** (FIG. 4). In one embodiment, the loop **232** may include a slip knot configured to tighten around the arm **146a** when the filament **230** is proximally retracted. In some embodiments, the filament **230** may be attachable either directly or indirectly to the first medical device **146** by adhesives, soldering, welding, brazing, or mechanical attachments such as bands, clamps, or ratchet, or combinations thereof. For example, the filament **230** may be looped around the first medical device **146** and set by applying an adhesive to the filament to itself. In some embodiments, the first medical device is provided with an aperture, and an end of the filament **230** is passed through the aperture and then expanded in diameter (e.g., by knotting, heat staking, having another element coupled to, etc., the end of the filament) to maintain the end of the filament in place. The handle **144** of the first working instrument **140** may then be actuated to move the arms **146a**, **146b** of the first medical device **146** from the first position to the second position and the first endoscopic instrument **140** may be proximally retracted such that the first medical device **146** is disposed (e.g., protected, shielded, hidden, loaded etc.) within the lumen **224** of the cap **220** and/or the working channel **214** of the elongate tubular member **210**. The endoscopic system **200**, with the first medical device **146** disposed within the lumen **224** and/or working channel **214** and the first filament **230** secured to the arm **146a**, may then be advanced through a body lumen of a patient and the distal end of the cap **220** positioned adjacent to a target tissue. The first endoscopic instrument **140** may then be distally advanced through the working channel **214** to position the first medical device **146** distally beyond the distal end of the cap **220**. The handle **144** may then be actuated to move the arms **146a**, **146b** of the first medical device **146** from the second position to the first position and the open arms **146a**, **146b** placed in contact with a surface of the target tissue. The handle **144** may then be actuated to move the arms **146a**, **146b** from the first position to the second position to engage (e.g., clamp) the first medical device **146** with the target tissue. In some embodiments, the handle **144** may be actuated to disengage the first medical device **146** from the

sheath **142** of the first endoscopic instrument **140**. In some embodiments, the handle **144** may actuate the instrument **140** to engage with the tissue prior to and/or during a procedure.

(20) In various embodiments, the first endoscopic instrument **140** may then be removed (e.g., withdrawn) from the working channel **214** of the elongate tubular member **210** and a second endoscopic instrument **150** may be advanced through the working channel to position a second medical device (e.g., an electrocautery knife, resection tool, etc.) attached to a distal end of the second endoscopic instrument **150** distally beyond the distal end of the cap **220** and adjacent to the target tissue **148**. As discussed above, a proximal end **236** of the first filament **230** may extend beyond the proximal end of the elongate tubular member **210** and external to the patient such that a medical professional may proximally retract the first filament **230** to apply tension/traction as the second medical device manipulates (e.g., dissects) the target tissue **148**.

(21) In one embodiment, the first medical device **146** may be engaged with the portion of the target tissue being dissected such that the target tissue may be retrieved from within the patient. Alternatively, the first medical device **146** may be engaged with the target tissue adjacent to the portion being excised. In such an embodiment, the physician or other medical professional may cut the first filament (e.g., using the second medical instrument) such that the first medical device **146** may remain within the body lumen after the medical procedure is completed. The portion of the target tissue engaged by the detached first medical device **146** may eventually separate or slough off and the first medical device **146** may be expelled/removed from within the patient by the body's natural course. Alternatively, the medical professional may proximally retract the first filament with sufficient force to either break (e.g., snap) the first filament or disengage (e.g., pull free) the arms of the first medical device from the target tissue.

(22) Referring to FIG. 5, in one embodiment, an endoscopic system **300** of the present disclosure may include the same or similar elements as endoscopic system **200**, and further include a second guide **227** (e.g., suture guide, filament guide, tubular support, ring, etc.) defining an open channel therethrough attached to/mounted on the outer surface of the cap **220**. Although the first and second guides **226**, **227** are depicted as positioned on substantially opposite sides of the cap **220**, in various embodiments the first and second guides **226**, **227** may be attached to/mounted on the outer surface of the cap **220** at a variety of longitudinal and/or axial locations relative to each other. In addition, or alternatively, additional guides (e.g., third, fourth, fifth guides, etc.) may be attached to/mounted on the outer surface of the cap **220** in a variety of different patterns and/or configurations relative to each other.

(23) In various embodiments, a second filament **231** (e.g., suture, wire, string, etc.) may be extendable along an outer surface of the elongate tubular member **210** and an outer surface of the cap **220** and through the second guide **227** such that a distal end **233** of the second filament **231** may attach to an arm **147a**, **147b** of a second medical device **147**. In use, and by way of example, the endoscopic system **300** may be withdrawn from the patient with the first filament **230** extending through the channel of the first guide **226** and the first medical device **146** engaged with the target tissue **148**. The second medical device **147** may then be attached to the distal end of the sheath **142** of the first endoscopic instrument **140**, an arm **147a**, **147b** of the second medical device attached to the second filament **231** and the second medical device **147** loaded within the lumen **224** of the cap **220** and/or the working channel **214** of the elongate tubular member **210** by following the steps outlined above.

(24) The endoscopic system **300**, with the second medical device **147** disposable within the lumen **224** and/or working channel **214** and the second filament **231** secured to one of the arms **147a**, **147b** of the second medical device **147**, may be reintroduced through the body lumen of the patient and the distal end of the cap **220** repositioned adjacent to the target tissue **148**. The medical professional may then actuate the handle **144** as discussed above to engage (e.g., clamp) a separate portion of the target tissue **148** with the second medical device **147** and then disengage/release the second medical device **147** by following the steps outlined above with respect to the endoscopic

system **200**.

(25) The first endoscopic instrument **140** may again be removed (e.g., withdrawn) from the working channel **214** of the elongate tubular member **210** and the second endoscopic instrument **150** introduced/reintroduced through the working channel to reposition the second medical device attached to a distal end of the second endoscopic instrument adjacent to the target tissue. The medical professional may then proximally retract the respective proximal ends **236**, **237** of the first and second filaments **230**, **231** as necessary to apply the desired amount (e.g., force) and/or direction of tension/traction as the second medical device manipulates (e.g., dissects) the target tissue **148**.

(26) In one embodiment, the first and second filaments **230**, **231** may include a distinguishing characteristic, such as a distinctive color code, such that the medical professional may visually identify which filament(s) to retract and/or release to properly position the target tissue for manipulation by the second medical device. For example, when viewing the target tissue through the elongate tubular member, the physician or other medical professional may identify the first filament **230** as a first color (e.g., blue) and the second filament **231** and a second color (e.g., red). The physician or other medical professional may then retract the proximal end of the first (e.g., blue) filament **230** to apply tension/retraction to the portion of the target tissue engaged with the first medical device **146** while allowing the portion of the target tissue engage with the second medical device **147** to remain slack. In addition, or alternatively, the first and second medical devices **146**, **147** may include a color which matches or otherwise corresponds to the respective colors of the first and second filaments **230**, **231** to which they are attached to further facilitate the medical professional's ability to apply and release tension/traction as necessary.

(27) Although the second filament **231** is described as being attached to the second medical device **147** by first removing the endoscopic system **300** from within the patient, in various embodiments the distal end of the second filament **231** may be attached to an arm **147a**, **147b** of the second medical device **147** within the body lumen, e.g., by pre-loading the second filament **231** along the outer surface of the elongate tubular member **210** and cap **220** and through the channel of the second guide **227**, then passing an arm of the second medical device through the loop **233** (e.g., slip-knot) of the second filament **231** within the body lumen and then proximally retracting the second filament **231** to tighten the loop **233**.

(28) Referring to FIG. **6**, in one embodiment, an endoscopic system **400** of the present disclosure may include the same or similar elements as endoscopic systems **200**, **300** and further include a flexible elongate shaft **234** extendable along an outer surface of the elongate tubular member **210** and cap **220** and through the channel of the first guide **226**. In various embodiments, the flexible elongate shaft **234** may be slidably and/or rotatably disposable within the first guide **226** such that a distal end **238** of the flexible elongate shaft **234** may be extended distally beyond a distal end of the cap **220**. A first filament **230** may be extendable through a full length of the flexible elongate shaft **234** such that a distal end **232** of the first filament **230** may be attached to an arm of a first medical device **146** and a proximal end **236** of the first filament **230** may extend outside the patient, as discussed above. In various embodiments, the flexible elongate shaft **234** may include sufficient rigidity (e.g., columnar strength) such that a medical professional may distally advance the flexible elongate shaft **234** through the channel of the first guide **226** to apply tension/traction to the first medical device **146** engaged with a target tissue **148** in a direction opposite to (e.g., distally away from) the distal end of the cap **220**.

(29) Although the endoscopic system **400** is described as including a single flexible elongate shaft **234** extendable along an outer surface of an endoscope that includes a cap **220** with a first guide **226** attached/mounted thereto, in various embodiments an endoscopic system of the present disclosure may include a cap with more than one guide attached/mounted thereto (e.g., first and second guides **226**, **227** of system **300**), each of which may slidably and/or rotatably receive separate flexible elongate shafts therethrough. In various additional embodiments, an endoscopic

system of the present disclosure that may include multiple guides (e.g., first and second guides **226**, **227** of system **300**) attached to/mounted on the outer surface of the cap **220** may include a flexible elongate shaft extending through one of the guides and a filament without a flexible elongate shaft extending through the other guide, e.g., to allow the physician to apply tension/traction in the proximal and distal directions relative to the distal end of the cap. In various additional embodiments, a flexible elongate shaft **234** of the present disclosure may be advanced over a first or second filament **230**, **231** of the present disclosure and through the respective channel of the first or second guides **226**, **227** while a medical device attached to the distal end of that filament is engaged with a target tissue, e.g., to allow the physician to switch the direction of tension/traction applied to the target tissue.

(30) Referring to FIG. 7, in one embodiment, a proximal end of a flexible elongate shaft **234** of the present disclosure may be attached to a handle assembly **138**. In various embodiments, the handle assembly **138** may include a first gripping element **138a** (e.g., thumb ring), a second gripping element **138b** (e.g., first finger ring) and a third gripping element **138c** (e.g., a second finger ring). The first gripping element **138a** may be attached to the proximal end of the flexible elongate shaft **234** and the second and third gripping elements **138b**, **138c** may be pivotably and/or slidably attached to the handle assembly **138** by a connecting arm **139**. A proximal end of a filament (e.g., first or second filaments **230**, **231**) of the present disclosure may be attached (e.g., tied, glued, etc.), either directly or indirectly, to the connecting arm **139** between the second and third gripping elements **138b**, **138c**. In one embodiment, a medical professional may hold the handle assembly **138** in their hand and proximally retract and/or distally advance the flexible elongate shaft **234** through the channel of a guide (e.g., first or second guides **226**, **227**) of the present disclosure using the first gripping element **138a**, e.g., to apply or release tension/traction of the filament in a distal direction relative to the distal end of the cap **220**. In addition, or alternatively, the medical professional may proximally retract and/or distally extend the filament through the guide by longitudinally pivoting and/or sliding the second and third gripping elements **138b**, **138c** relative to the handle assembly **138** (e.g., in a back-and-forth direction). In various embodiments, the first gripping element **138a** may provide coarse/gross control of the flexible elongate shaft **234**, e.g., to position or reposition the distal end **238** of the flexible elongate shaft relative to the target tissue, and the second and third gripping elements **138b**, **138c** may provide fine-tune/incremental control of the filament to precisely manipulate the target tissue.

(31) Referring to FIG. 8, in one embodiment, a handle assembly **138** of the present disclosure may include the same or similar elements as the handle assembly of FIG. 8, but with second and third gripping elements **138b**, **138c** that are independently actuatable (e.g., not attached by a connecting arm, etc.). In one embodiment, the second gripping element **138b** may be pivotably and/or slidably attached to a proximal end of a filament (e.g., first or second filaments **230**, **231**) of the present disclosure and the third gripping element **138c** may be pivotably and/or slidably attached to a proximal end of the flexible elongate shaft **234**. In various embodiments, the first gripping element **138a** may provide coarse/gross control of the flexible elongate shaft **234** (as discussed above), and the second and third gripping elements **138b**, **138c** may provide fine-tune/incremental control of the filament and flexible elongate shaft independent of each other. For example, the second gripping element **138b** may be actuated to move the filament relative to (e.g., in a back-and-forth direction) the flexible elongate shaft and/or the third gripping element **138c** may be actuated to move the flexible elongate shaft relative to (e.g., in a back-and-forth direction along a longitudinal axis) the filament.

(32) Although the first and second guides **226**, **227** of the present disclosure are depicted as positioned adjacent to a proximal end of the cap **220**, in various embodiments the first and second guides **226**, **227** may be attached to/mounted on the outer surface of the cap **220** at a variety of locations (e.g., a mid-point of the cap, a proximal end of the cap, etc.). In addition, or alternatively, the first and second guides **226**, **227** may extend a full length of the cap **220** and/or extend distally

beyond the distal end of the cap **220**.

(33) In addition, although first and second guides of the present disclosure are described as associated with (e.g., attached to, mounted on) an outer surface of a cap **220**, and the above-described embodiments are described with reference to a cap element, it will be appreciated that the principles of the above disclosure are applicable to each embodiment even if a cap element is not provided. As such, references to a cap on a structure should be understood to include references to an element which is a part of (unitary or separate) the distal end of the structure. For instance, in various embodiments the guides may be provided along or attached to/mounted on an outer surface of the flexible elongate tubular member either with or without a cap **220** attached thereto. As above, the guide(s) may be distributed along the outer surface of the flexible elongate tubular member in a variety of different patterns, orientations and/or numbers, such as those described with reference to a cap. For example, a series of guides may be disposed along an outer surface of the flexible elongate tubular member in evenly or unevenly spaced intervals along a longitudinal or non-longitudinal axis thereof. In some embodiments, the guides may extend along an entire outer surface of the flexible elongate tubular member and/or the cap, e.g., to provide a contiguous enclosed channel through which the filament(s) may extend. In some embodiments, the guides may include one or more straps or bands disposed around various circumferential portions of the outer surface of the flexible elongate tubular member and/or cap. Additionally or alternatively, a flexible elongate shaft is extendable along an outer surface of the elongate tubular member regardless of whether a cap is present. In some embodiments, a flexible elongate shaft is extendable through a guide extending along or mounted on the outer surface of an elongate tubular member. A filament may extend through any of the above embodiments, such as through the guide and/or flexible elongate shaft of any of the above embodiments.

(34) In various embodiments, the second medical device of the present disclosure is not limited to a tissue cutting element (e.g., electrocautery knife, etc.), but may include a variety of medical instruments configured to manipulate a target tissue (e.g., ablative elements, needles or syringes configured to inject agents into the target tissue, etc.).

(35) In various additional embodiments, the first and second medical devices **146**, **147** of the present disclosure herein may include a variety of hemostasis and non-hemostasis tissue clips configured to secure/engage the first and second filaments **230**, **231** to the target tissue. For example, a tissue clip contemplated for use with the disclosed endoscopic systems **200**, **300**, **400** may include a naturally open/biased configuration configured to move to a closed/clamped configuration upon actuation by a handle assembly. In addition, or alternatively, a tissue clip contemplated for use with the disclosed tissue retraction/traction device may include a naturally closed/biased configuration configured to move an open configuration upon actuation by a handle assembly. In addition, or alternatively, fasteners other than the described tissue clips may be used to secure/engage the first and second attachment members of the disclosed tissue retraction/traction device to the wall of a body lumen. Examples of fasteners may include, but are not limited to, those described in U.S. Pat. No. 10,952,717, issued May 13, 2020, and titled “Tissue Traction Bands and Methods of Use Thereof”; U.S. Patent Application Publication No. 2020/0129181, filed Oct. 30, 2019, and titled “Clip Devices, Systems, and Methods for Engaging Tissue”; U.S. Pat. No. 11,147,564, issued May 13, 2020, and titled “Tissue Clip Devices, Systems, and Traction Methods”; U.S. Patent Application Publication No. 2018/0263614, filed Mar. 19, 2018, and titled “Tissue Retraction Device and Delivery System”; and U.S. Pat. No. 8,062,311, issued Nov. 22, 2011, and titled “Endoscopic Hemostatic Clipping Apparatus”, all of which are herein incorporated by reference in their entireties for all purposes. Other features and aspects of these patents and patent applications, as well as U.S. Patent Application Publication No. 2020/0360006, filed on May 13, 2020, and titled “Tissue Traction Bands And Methods For Tissue Traction”, which application is herein incorporated by reference in its entirety for all purposes, may complement devices and methods of the present disclosure and may be used therewith.

(36) In addition, it should be appreciated that the first and second medical devices **146, 147** of the present disclosure are not necessarily drawn to scale, but may be represented in a somewhat enlarged configuration to provide the requisite level of detail to understand and practice the embodiments of the present disclosure.

(37) All of the devices and/or methods disclosed and claimed herein can be made and executed without undue experimentation in light of the present disclosure. While the devices and methods of this disclosure have been described in terms of preferred embodiments, it may be apparent to those of skill in the art that variations can be applied to the devices and/or methods and in the steps or in the sequence of steps of the method described herein without departing from the concept, spirit and scope of the disclosure. All such similar substitutes and modifications apparent to those skilled in the art are deemed to be within the spirit, scope and concept of the disclosure as defined by the appended claims.

Claims

1. A system for use with a flexible elongate tubular member extendible within and through a patient and having a working channel therethrough, the system comprising: a first guide mounted with respect to an outer surface of the flexible elongate tubular member; and a first filament extendable through the first guide and along an outer surface of the flexible elongate tubular member; wherein the first filament has a distal end configured to be coupled to tissue via an instrument extended through the working channel of the flexible elongate tubular member, and a proximal end extendable proximally to outside the patient for control by a medical professional to proximally pull on the first filament to apply traction to the tissue.
2. The system of claim 1, wherein the first filament extends through the first guide without an additional element therearound.
3. The system of claim 1, wherein the first filament is extendable proximally outside the patient to be retractable by a physician.
4. The system of claim 1, further comprising: a second guide attached to an outer surface of the cap; and a second filament extendable along the outer surface of the flexible elongate tubular member.
5. The system of claim 4, further comprising a flexible elongate shaft extendable along the outer surface of the flexible elongate tubular member and the cap and through the second guide, wherein the second filament is extendable through the flexible elongate shaft.
6. The system of claim 1, wherein the flexible elongate tubular member is an endoscope.
7. The system of claim 5, further comprising a first endoscopic instrument extending through the working channel of the flexible elongate tubular member.
8. The system of claim 6, wherein the first filament is extendable from a first end attached to a distal end of the first endoscopic instrument to a second end extended outside the patient and retractable by a physician.
9. A system for use with a flexible elongate tubular member extendible within and through a patient and having a working channel therethrough, the system comprising: a first guide extendable along at least a portion of an outer surface of the flexible elongate tubular member; a first filament having a distal end and a proximal end and extendable through the first guide; and a first instrument extendable through the working channel of the flexible elongate tubular member and having a distal end configured to be coupled with the distal end of the filament to couple the distal end of the first filament with respect to tissue within the patient.
10. The system of claim 9, wherein the first instrument includes first and second arms, the distal end of the first filament disposed around one of the arms.
11. The system of claim 10, wherein the first and second arms are movable with respect to each other to engage target tissue to couple the distal end of the filament to the target tissue.

12. The system of claim 9, wherein the distal end of the first filament is attached to the distal end of the first instrument by adhesives, soldering, welding, brazing, or mechanical attachments such as bands, clamps, or ratchet, or combinations thereof.
13. The system of claim 9, wherein the first instrument comprises a sheath and a first medical device coupled to a distal end of the sheath at the distal end of the first instrument.
14. The system of claim 9, further comprising a second guide extending along a portion of the outer surface of the endoscope and a second filament extending through the second guide and along an outer surface of the endoscope.
15. The system of claim 14, further comprising a flexible elongate shaft extendable along the outer surface of the endoscope and through the second guide, wherein the second filament extends through the flexible elongate shaft.
16. The system of claim 9, wherein the proximal end of the filament is extendable proximally to outside the patient for control by a medical professional to proximally pull on the first filament to apply traction to the tissue via the distal end of the filament and the distal end of the first instrument.
17. A method of assembling a system for performing an endoscopic procedure within a patient, the method comprising: advancing a first instrument through a working channel of an elongate tubular member to advance a distal end of the first instrument distal to the distal end of the elongate tubular member; advancing a first filament along an outer surface of a flexible elongate tubular member to a distal end of the flexible elongate tubular member; and coupling a distal end of the first filament to the distal end of the first instrument so that the system is ready to be advanced through a body lumen of a patient to position the distal end of the elongate tubular member adjacent a target tissue to couple the distal end of the first filament, via the distal end of the first instrument, to the target tissue so that pulling proximally on the proximal end of the filament applies traction to the target tissue.
18. The method of claim 17, further comprising extending the first filament through a first guide mounted on an outer surface of the flexible elongate tubular member.
19. The method of claim 18, further comprising extending the first filament directly through the first guide without an additional element therearound.
20. The method of claim 17, wherein the target tissue is within a patient, the method further comprising extending the proximal end of the first filament outside the patient.
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